

Article

Effect of Layer Thickness and Scanning Parameters on Melt Pool Geometry and Track Continuity in Powder-Bed Arc Additive Manufacturing

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Abstract

Powder-bed arc additive manufacturing (PBAAM) may reduce the cost of powder-bed metal additive manufacturing and enable thicker layers than laser powder bed fusion (LPBF), but melt-track stability limits are not well established. Here, 316L stainless steel powder (15–53 μm) was melted by a TIG-based arc in a custom powder-bed system while varying current, travel speed, layer thickness and hatch distance. Single tracks on an inclined bed (≈ 0 –0.4 mm thickness) were used to identify continuity loss and melt-pool width, quantified from top-view images via width profiles, a gap-based continuity metric and the coefficient of variation. Parallel-track tests at 0.15, 0.20 and 0.25 mm layer thickness with hatch distances set to 25%, 50% and 75% of the measured melt-pool width assessed inter-track bonding and lack of fusion, and selected parameters were validated in five-layer builds. Higher current with low-to-moderate travel speeds produced wider, more stable melt pools on the inclined bed. Hatch ratios of 25–50% were the most effective for sustaining fusion in single layers and multi-layer builds, whereas 75% promoted unbonded regions and narrow-track morphologies. Overall, PBAAM can process substantially thicker layers with relatively simple equipment, but requires a narrow, carefully tuned window to balance continuity, fusion and heat accumulation.

Keywords: PBAAM; melt pool geometry; track continuity; layer thickness; hatch distance; 316L stainless steel

1. Introduction

Metal additive manufacturing (AM) technologies have, in recent years, been widely adopted in the aerospace, automotive, biomedical and energy sectors to produce complex-geometry, high-performance components. Among AM technologies, metal PBF processes distinguish themselves by their high dimensional accuracy and ability to manufacture intricate geometries. Traditionally, high-intensity energy sources such as lasers and electron beams are preferred in these processes [1].

In the literature, these methods are commonly referred to as selective laser melting (SLM) and electron beam melting (EBM). While they offer high accuracy and repeatability for complex geometries, they also suffer from drawbacks such as high equipment cost and limitations in productivity [1,2]. Although both technologies provide micron-level precision, the layer thickness is typically restricted to the range of 30–150 μm [2]. These constraints significantly increase build time and cost, especially for large-volume parts [3,4].



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Arc-energy-based powder bed additive manufacturing, here referred to as powder-PBAAM, has emerged as a lower-cost and higher-efficiency alternative to laser- and electron-beam-based systems [5]. Using an electric arc as the energy source enables the process to be implemented with relatively simple and accessible welding equipment, while the use of thicker layers offers the potential for higher build rates [6]. A key advantage of systems that employ an arc as the heat source is their ability to achieve very high deposition rates and to fabricate meter-scale, near-net-shape components [7,8]. However, since the melt pool in these methods is significantly larger than in laser-based systems, process precision and spatial resolution are reduced [9]. Consequently, achieving an optimum balance between productivity and accuracy is a central research challenge in arc-based systems. While laser PBF processes are characterized by high cooling rates and small melt pools, arc-based melting is associated with slower cooling and larger melt pool geometries [8,10]. This directly affects the resulting microstructure and dimensional accuracy. On the other hand, the ability to melt relatively thick layers (~0.3–0.4 mm and above) in a single pass with an arc-based energy input offers a significant advantage in terms of build time for large-volume parts [11].

Austenitic stainless steel 316L is one of the most widely used alloys in metal additive manufacturing due to its high toughness, corrosion resistance and good weldability [12]. Its good weldability and corrosion resistance make 316L suitable for a wide range of heat input regimes in AM, and it has repeatedly been selected in the literature as the benchmark alloy for LPBF, Laser Metal Deposition (LMD), wire arc additive manufacturing (WAAM) and hybrid processes [13–15]. The behaviour of 316L powder in laser-based PBF has been extensively studied, providing a rich database for comparison. Therefore, the use of this alloy in the present work allows the results of the new arc-based method to be benchmarked against well-established LPBF findings [4]. In addition, 316L is a steel that is readily melted by an arc and has a low susceptibility to cracking, which makes it a suitable choice for ensuring process stability in experimental studies [16].

The existing literature provides comprehensive data on WAAM and hybrid laser–arc systems; however, there is a distinct lack of detailed experimental characterization studies in which the arc energy source is applied directly to a powder bed. As a result, there is little information available on process maps, layer-by-layer melting dynamics and powder-melting efficiency for processes where an arc acts on a powder bed. This constitutes a significant research gap.

The aim of this study is to systematically investigate the influence of process parameters on melt pool geometry and multi-layer build formation PBAAM, where an electric arc is used as the energy source in a powder bed configuration. Three different experimental approaches are employed. In the first stage, single-track formation behaviour—widely accepted as the starting point of process window development in layer-wise AM—is examined. For this purpose, an inclined powder bed method is used to study the effect of continuously varying layer thickness on single-track formation. By enabling a gradual change in layer thickness along the track, the inclined bed approach allows the identification of optimum process windows. In this stage, the effects of layer thickness on track continuity and melt pool width and depth are observed for different current levels and travel speeds.

In the second stage, the parameters identified as suitable for single-track formation in the inclined bed experiments are used to investigate the capability of forming continuous surfaces through the merging of parallel tracks within a single layer. Two parallel tracks are produced on the powder bed at three different layer thicknesses (0.15 mm, 0.20 mm and 0.25 mm) and three hatch distances corresponding to 25%, 50% and 75% overlap.

These experiments are used to evaluate the fusion quality between adjacent tracks, surface smoothness and the occurrence of defects such as lack of fusion.

In the final stage, following the evaluation of single-track and intra-layer (hatch spacing) results, three-dimensional builds are produced to investigate multi-layer PBAAM. In this stage, the melt pool geometry, solidification morphology and dimensional accuracy are examined for multi-layer deposition obtained by successively stacking arc-melted powder layers. Layer height control, surface quality and structural integrity are evaluated for multi-layer builds. The findings are used to address existing gaps in the literature regarding the optimization of low-cost, arc-based powder bed manufacturing technologies. In this way, the study provides a basis for the optimization of process parameters, melt pool control, and the development of single- and multi-layer build strategies in arc-based powder bed fusion systems.

2. Materials and Methods

In this study, arc-based melting experiments were carried out on an additively manufactured powder bed using 316L stainless steel powder under different current, travel speed, layer thickness and hatch distance conditions within PBAAM framework.

2.1. Powder and Substrate

The feedstock was a 316L stainless steel powder with spherical morphology. The 316L stainless steel powder was supplied by Jiangsu Vilory Advanced Materials Technology Co., Ltd., Xuzhou, China (batch no. 202306-PRD-67). The powder was produced via gas atomization and had a particle size range of 15–53 μm . According to the supplier's quality certificate, the chemical composition (wt.%) was Cr: 16.83, Ni: 10.93, Mo: 2.27, Mn: 0.67, Si: 0.66, P: 0.033, C: 0.0080, S: 0.005 and Fe: balance. The oxygen and nitrogen contents were 510 ppm and 936 ppm, respectively. The particle size distribution values were D10: 18.7 μm , D50: 31.1 μm and D90: 51.7 μm . This size range is typical of powders used in LPBF and was considered suitable for forming a fluid melt pool under arc-based melting conditions. As substrate material, 2 mm thick 316L stainless steel plates of the same alloy were used. The plates were cut to dimensions of 140 mm $\times$ 10 mm, 30 mm $\times$ 10 mm and 30 mm $\times$ 30 mm to match the different experimental configurations and were placed in the recesses of a powder bed carrier. Before powder spreading, all substrate surfaces were ground and cleaned to promote good wetting and adhesion of the melt pool.

2.2. PBAAM System

The PBAAM system was built by integrating a gas tungsten arc welding power source with a three-axis motion platform. A schematic representation is given in Figure 1. Motion in the X–Y–Z axes was provided by stepper motors with high positioning accuracy, enabling precise adjustment of the electrode position and layer thickness.

A 1.6 mm diameter lanthanated tungsten electrode (2% La_2O_3) was mounted in the welding torch. The electrode tip was grounded so that the tip plane was parallel to the workpiece surface, to obtain a focused and stable arc on the powder bed. High-purity argon shielding gas was supplied through a gas lens at a flow rate of 6 L/min to ensure laminar flow and to minimize the influence of ambient air on the arc. A 220 A AC/DC TIG power source was used, operated in direct current (DC) mode for all experiments.

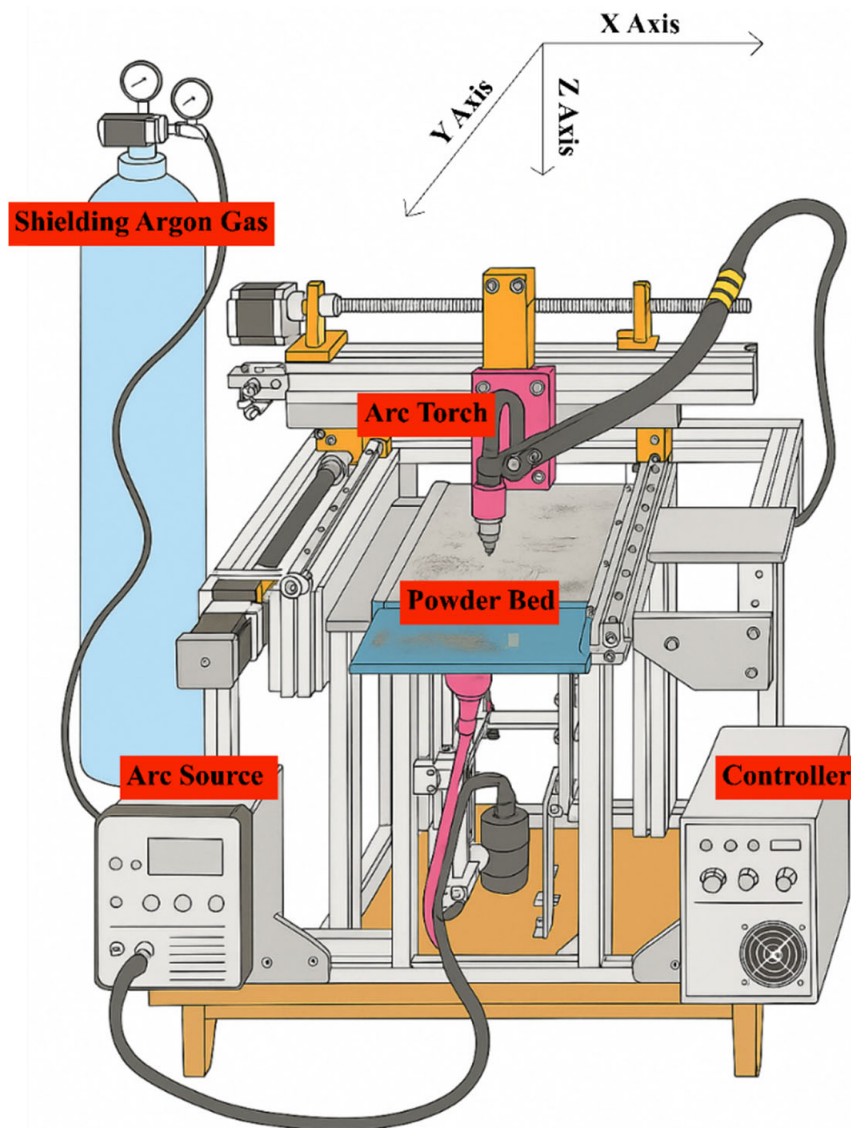


Figure 1. Schematic representation of PBAAM system.

2.3. Inclined-Bed Single-Track Experiments

To investigate single-track formation at different layer thicknesses, inclined powder bed experiments were conducted. For this purpose, one end of the substrate plate was elevated relative to the other before powder spreading, creating a linearly varying layer thickness along the scan direction (Figure 2). After spreading the 316L powder, the layer thickness was approximately 0 mm at the start of the track and increased up to about 0.4 mm at the end over a length of 140 mm.

For each experiment, a single melt track was produced along the inclined bed by translating the arc from the thin layer region toward the thicker region. Experiments were performed at three current levels (20, 25 and 30 A) and four travel speeds (2, 3, 4 and 5 mm/s). The selected parameter levels were not arbitrarily defined. They were established based on a prior systematic parameter-mapping study conducted under the same PBAAM configuration (identical torch setup, shielding conditions and electrode–powder-bed distance) [17]. In that preliminary investigation, the transition boundaries between lack-of-fusion behaviour, geometrically stable melting and excessive melt-pool spreading were experimentally identified. Currents below 20 A were insufficient to sustain continuous melting, whereas currents above 30 A promoted unstable melt-pool expansion. Similarly, travel speeds above 5 mm/s resulted in severe discontinuities, while speeds below

2 mm/s led to excessive heat accumulation and geometric overgrowth. The present study therefore focuses on a refined operational window established through prior experimental screening. After melting, visual inspection was used to determine the position where track continuity was lost. Cross-sections taken from selected positions along the track were prepared metallographically to measure melt pool width and depth.

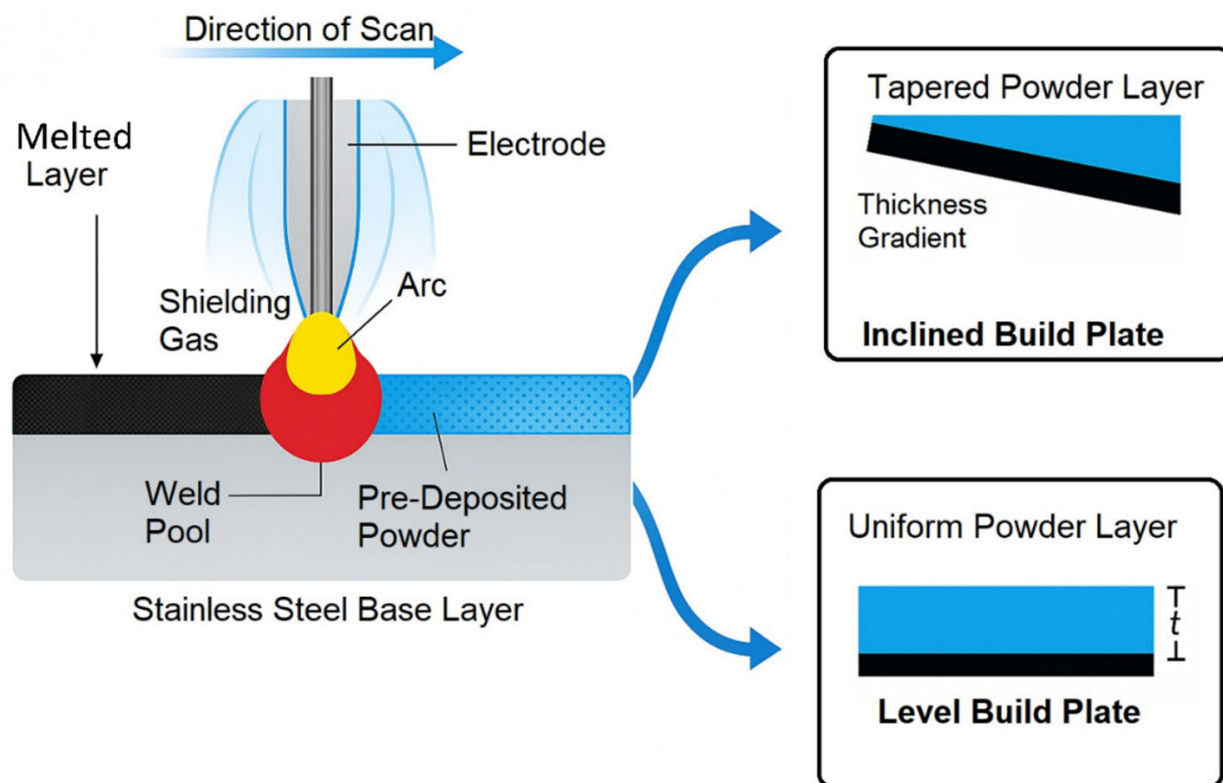


Figure 2. Schematic illustration of the inclined powder bed geometry and single-track melting configuration in PBAAM.

2.4. Hatch Spacing Experiments (Parallel Tracks)

Using the inclined-bed results, combinations of current and travel speed that produced continuous single tracks were identified as candidate parameters for multi-track experiments. In this second stage, powder was spread on 10 mm × 30 mm substrates at nominal layer thicknesses of 0.15, 0.20 and 0.25 mm. For each layer thickness, two parallel tracks were produced in the powder bed. The hatch distance between the centers of the two tracks, h , was selected as 25%, 50% and 75% of the melt pool width w_{sp} measured in the corresponding single-track condition, i.e.:

$$h = 0.25 w_{sp}, 0.50 w_{sp}, 0.75 w_{sp} \quad (1)$$

These choices correspond to nominal overlaps of 75%, 50% and 25%, respectively. After melting, the continuity of the tracks, the fusion in the overlap region and the solidification morphology were evaluated. Cross-sections were used to inspect whether adjacent melt pools fully merged, the penetration depth into the substrate, and the presence of lack-of-fusion defects.

2.5. Multi-Layer PBAAM Experiments

In the third stage, multi-layer builds were produced using parameter sets that had shown good fusion in single-layer tests. For each selected combination of layer thickness and process parameters, five layers were deposited on 30 mm × 30 mm substrates using

hatch distances corresponding to 25% and 50% overlap. Cross-sections taken at intermediate heights were examined to characterize melt pool geometry, interlayer fusion, layer height stability and geometric accuracy.

2.6. Optical and Thermal Imaging

All melting experiments were recorded using a Sony Mark 5 high-speed camera (Sony Corporation, Minato-ku, Tokyo, Japan). To mitigate the strong arc radiation and obtain clear images of both the arc and the melt pool, a neutral density (ND) filter was used, and the following settings were applied: ISO 80, aperture f 2.8 and shutter speed $1/12,000$ s. In addition, a Testo 865 infrared camera (Testo SE & Co. KGaA, Titisee-Neustadt, Germany) was employed to record thermal images of the process and to analyze the temperature distribution in the powder bed during melting.

2.7. Metallographic Preparation and Melt Pool Imaging

To reveal the melt pool geometry and the bonding at the powder–substrate interface, samples were sectioned using a precision cutting machine (MICRACUT 200-S, Metkon Instruments Inc., Bursa, Turkey), mounted in epoxy, and then ground and polished. Metallographic observations were conducted using an Optimus optical microscope (AOB TEST Sistemleri/AOBLAB, Istanbul, Türkiye) equipped with a $10\times$ objective and a UCMOS digital camera (Hangzhou ToupTek Photonics Co., Ltd., Hangzhou, Zhejiang, China). Grinding was performed with 800, 1000, 1200 and 2500 grit SiC papers, followed by polishing with diamond suspension on a cloth. The melt pool boundaries were made visible by etching with Viella's reagent (5 mL HCl, 1 g picric acid, 100 mL 95% ethanol).

2.8. Single-Track Image Processing and Continuity Analysis

Top-view images were analyzed using a five-stage image processing pipeline (Figure 3) to assess geometric continuity and width stability. Top-view colour images of single tracks were recorded and imported into MATLAB 2019a (MathWorks, Inc., Natick, MA, USA). Each RGB image I_{RGB} was converted to a grey-scale image $I_g(x, y)$ by a standard luminance-preserving mapping [18].

$$I_g(x, y) = f(I_{RGB}(x, y)) \quad (2)$$

Binary masks were obtained via Otsu thresholding, followed by morphological operations (dilation $\rightarrow$ erosion) to smooth contours [19]:

$$M(x, y) = \begin{cases} 1, & I_g(x, y) \geq T, \\ 0, & I_g(x, y) < T, \end{cases} \quad (3)$$

Local track width was measured along the centerline [20]:

$$w(s_i) = x_{\text{right}}(s_i) - x_{\text{left}}(s_i) \quad (4)$$

Gap Test: The first and last 1–2 mm were excluded to eliminate arc initiation/extinction [20] transients. Total gap length where $w(s_i) < w_{\min}$

$$L_{\text{gap}} = \sum_{k=1}^N \Delta s_k \quad (5)$$

The threshold $L_{\text{crit}} = 0.5$ (corresponding to 15–25% of typical track width) was established through systematic validation. The track is classified as

$$\begin{aligned} L_{\text{gap}} > 0.5 \text{ mm} &\Rightarrow \text{Discontinuous (gap – dominated) track,} \\ L_{\text{gap}} \leq 0.5 \text{ mm} &\Rightarrow \text{Continuous in the gap sense.} \end{aligned} \quad (6)$$

Uniformity test (width stability): While the gap test focuses on local interruptions, the uniformity test quantifies global width variability along the entire track. Large fluctuations are associated with unstable melt-pool dynamics and degraded surface quality.

$$CV_w = \frac{\sigma_w}{\mu_w} \quad (7)$$

Based on preliminary PBAAM experiments in which manual visual labels (uniform/non-uniform) were compared with computed CV_w values, the selected threshold is:

$$CV_{crit} = 0.12 \quad (8)$$

The threshold $CV_{crit} = 0.12$ was determined by scanning the range 0.10–0.15 [21] and selecting the value yielding the highest agreement (~97%) with track quality labels.

$$\begin{cases} CV_w \leq 0.12 \Rightarrow \text{Stable (uniform) track} \\ CV_w > 0.12 \Rightarrow \text{Unstable (non-uniform) track} \end{cases} \quad (9)$$

Final output of the framework: Each track is assigned two labels: (i) continuous/discontinuous based on L_{gap} , and (ii) stable/unstable based on CV_w . As discussed in the validation section, these two labels are used jointly to assess track quality.

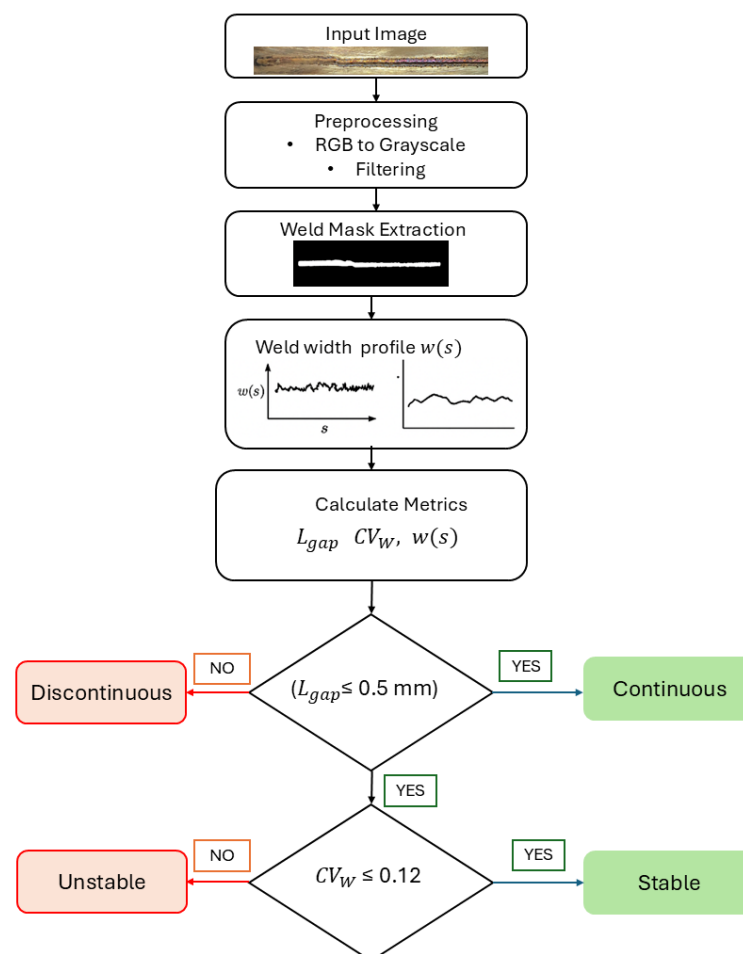


Figure 3. Flowchart of the image processing procedure used for single-track analysis.

2.9. Inter-Track Bonding Analysis

Because hatch distance (HD) directly controls the overlap between neighbouring tracks, lack-of-fusion (LoF) defects often initiate in regions with insufficient inter-track bonding. In addition to single-track continuity, the geometrical relationship between adjacent tracks was therefore evaluated using an inter-track gap/overlap metric and a connected-length ratio (CL). Similar gap/overlap-based indicators have been used to study LoF formation under varying hatch spacing in LPBF [21].

2.9.1. Inter-Track Gap/Overlap

For each pair of neighbouring tracks, the segmentation and boundary-extraction procedure of Section 2.8. was applied to obtain the left and right boundaries of the first track $x_{\text{left}}^{(1)}(s_i)$ and $x_{\text{right}}^{(1)}(s_i)$, and of the second track, $x_{\text{left}}^{(2)}(s_i)$ and $x_{\text{right}}^{(2)}(s_i)$. The inter-track distance is defined as

$$g(s_i) = x_{\text{left}}^{(2)}(s_i) - x_{\text{right}}^{(1)}(s_i). \quad (10)$$

By convention:

- $g(s_i) > 0$: a positive inter-track gap exists; powder or base material remains unmelted between the tracks, indicating a strong LoF risk.
- $g(s_i) \leq 0$: the two molten pools geometrically overlap or at least meet; adequate metallurgical bonding is expected in this region.

This contour-based gap/overlap formulation is consistent with geometric LoF criteria used to link melt-pool width, hatch spacing and porosity in LPBF [21].

2.9.2. Connected-Length Ratio (CL)

To condense the bonding behaviour between two neighbouring tracks into a single scalar quantity, the connected-length ratio CL is defined. First, an indicator function

$$I(g(s_i) \leq 0) = \begin{cases} 1, & g(s_i) \leq 0, \\ 0, & g(s_i) > 0 \end{cases} \quad (11)$$

is introduced. The connected-length ratio is then

$$CL = \frac{\sum_{i=1}^n I(g(s_i) \leq 0) \Delta s}{L_{\text{total}}}, \quad (12)$$

where Δs is the physical spacing between consecutive samples along the scan path and L_{total} is the total analysed length. By construction, $0 \leq CL \leq 1$:

$CL \approx 1$ indicates that almost the entire length analysed exhibits overlap between neighbouring tracks; inter-track bonding is excellent and the LoF risk is low.

$CL \ll 1$ indicates that large fractions of the inter-track region remain unbonded; this behaviour is typically observed at large hatch distances or under-melted conditions and correlates with high LoF pore density [21].

In the present PBAAM study, CL is evaluated for each parameter set and plotted against hatch distance to identify regimes with acceptable bonding. Combined with the single-track continuity criteria in Section 2.8, this yields a compact, image-based quality map for the inclined powder bed.

Although one physical track was produced per parameter set, the width profile extraction yields several hundred independent spatial samples along each track. Statistical descriptors such as mean width, standard deviation and coefficient of variation are therefore computed over a large spatial dataset rather than a single scalar value. The experimental setup operates under deterministic arc conditions (constant current, travel speed, shielding

gas flow and substrate inclination), minimizing stochastic variability. Consequently, the reported continuity metrics possess strong internal statistical robustness.

3. Results

3.1. Single-Track Melting on the Inclined Powder Bed

The influence of current and travel speed on single-track PBAAM behaviour was first evaluated on the inclined powder bed. For all parameter sets listed in Section 2, the resulting melt tracks were examined both macroscopically (solidified appearance, balling, breakup) and quantitatively using the continuity and uniformity metrics defined in Section 2.8 (gap length L_{gap} , median width $\tilde{w}$, minimum width w_{min} , total gap length within each segment, and the width statistics μ_w, σ_w, CV_w).

Visual inspection of the solidified tracks showed that melting does not start exactly at the geometric beginning of the inclined bed. At the leading edge, columnar powder accumulations at the sides and bottom of the recess disturb the initial arc behaviour, so that a stable melt pool is not formed in this region. To avoid this artefact, the electrode was positioned slightly inboard of the inclined-bed edge such that the arc footprint did not intersect the powder piles at the boundary.

Thermographic images acquired at the end of each track (Figure 4) reveal a clear increase in overall temperature level and heat accumulation with increasing current. As the current rises from 20 A to 30 A, the number of small, isolated solidified beads around the main track decreases, whereas their size and the extent of a continuous heat-affected band increase, indicating a more energetic and continuous melt pool. In contrast, for 20 A and 25 A at higher travel speeds the available line energy is insufficient, especially towards the thick-powder end of the inclined bed. Under these low-energy conditions, the track loses its linearity: the melt pool periodically collapses into spherical agglomerates, and balling is observed within the arc-affected zone.

Beyond these qualitative observations, the inclined-bed tracks produced with a 20–30 A current and a 2–5 mm/s travel speed were analysed segment-wise using the procedure of Section 2.8. Each track was automatically divided into continuity segments, and for each segment the metrics L_{gap} , $\tilde{w}$, w_{min} and CV_w were computed. The numerical results are summarised in Table 1. Segment-based evaluation shows that the quality behaviour along a single PBAAM track is too heterogeneous to be described by a single global indicator such as the average width or one overall CV_w value. Therefore, continuity and uniformity metrics were evaluated within the active continuity interval, excluding ignition and extinction transients. Additionally, the spatial extent of the stable region was quantified and reported separately to capture thickness-dependent process stability. In every specimen the track is composed of several segments, each formed under different local thermal, geometric and flow conditions, and therefore each requires individual quality assessment.

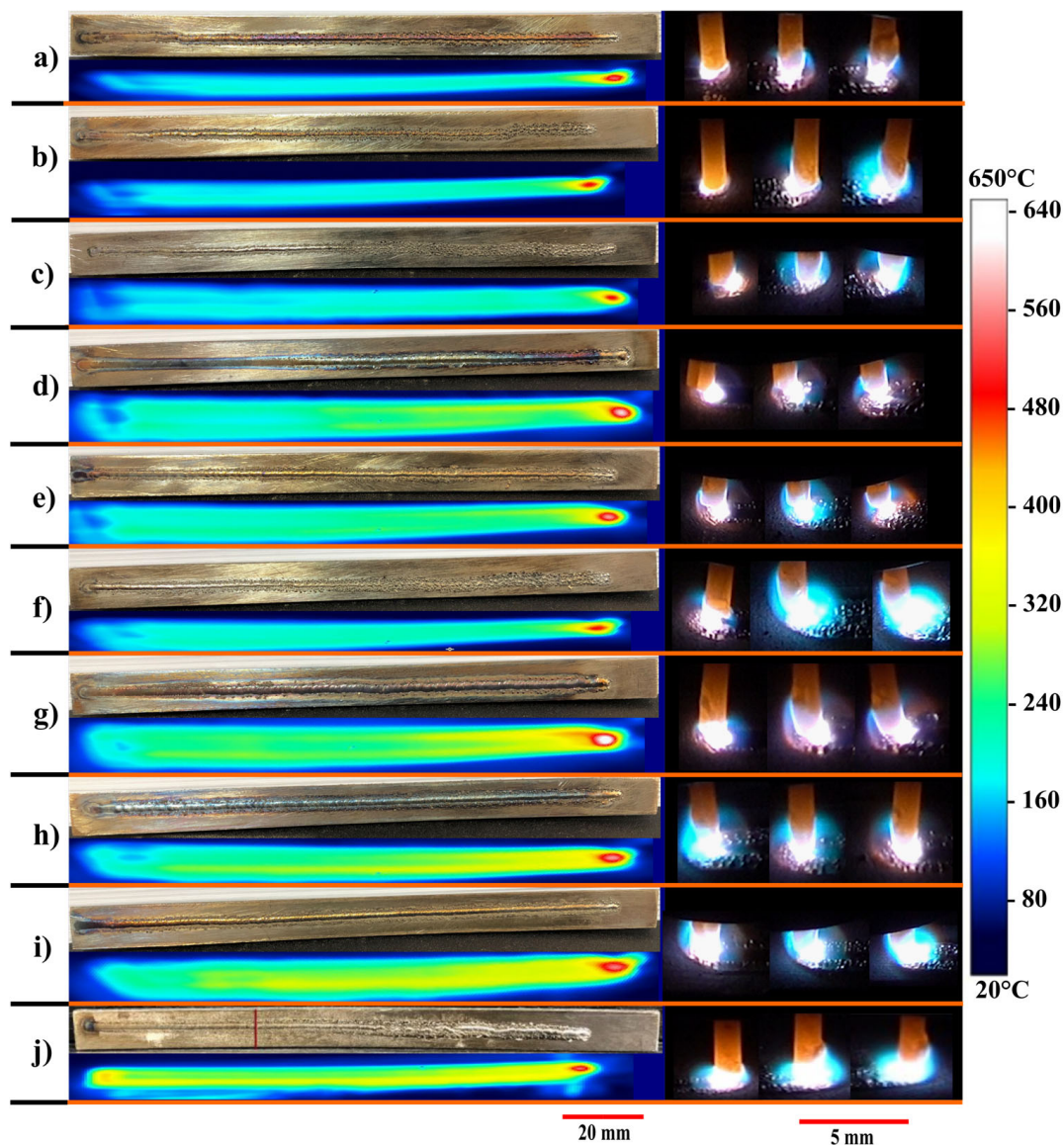


Figure 4. Solidified single tracks produced on the inclined powder bed under different combinations of current (20–30 A) and travel speed (2–5 mm/s). For 20 A: (a) 2 mm/s, (b) 3 mm/s, and (c) 4 mm/s; for 25 A: (d) 2 mm/s, (e) 3 mm/s, and (f) 4 mm/s; for 30 A: (g) 2 mm/s, (h) 3 mm/s, (i) 4 mm/s, and (j) 5 mm/s. Representative arc images at 35, 70, and 105 mm along the inclined bed and post-termination thermal views are also shown.

Table 1. Continuity assessment within the active interval and spatial stability limits for inclined-bed single tracks.

Sample	Continuous Length (mm)	Continuous Fraction (%)	L_{gap} (mm)	CV_w	Track-Level Class
20 A, 2 mm/s	12.404	88.6	15.4	0.156	Unstable
20 A, 3 mm/s	11.480	82.0	22.1	0.612	Critical
20 A, 4 mm/s	8.918	63.7	42.3	0.266	Critical
25 A, 2 mm/s	12.642	90.3	12.5	0.138	Unstable
25 A, 3 mm/s	11.774	84.1	18.0	0.207	Unstable
25 A, 4 mm/s	9.716	69.4	35.6	0.297	Critical
30 A, 2 mm/s	13.650	97.5	4.2	0.082	Stable
30 A, 3 mm/s	13.412	95.8	6.1	0.109	Stable
30 A, 4 mm/s	13.090	93.5	8.9	0.124	Unstable
30 A, 5 mm/s	10.360	74.0	25.6	0.292	Critical

3.1.1. Gap Analysis

The gap analysis indicates that the primary cause of segmentation is local melt-pool instability. Segments with low gap length ($L_{\text{gap}} \leq 10$ mm) are mainly observed at high current (30 A) and low-to-moderate travel speeds (2–3 mm/s). In these segments the melt pool propagates in a relatively stable manner and the track remains as a single long continuous region. In contrast, segments with medium to high gap lengths ($L_{\text{gap}} > 20$ mm) occur when the travel speed is increased or the current is reduced. Under such low line-energy conditions, the melt-pool continuity breaks down, the track is fragmented into several short segments, and the overall continuity is lost.

3.1.2. Uniformity Analysis

The uniformity metric CV_w plays a key role in describing the internal quality of each segment. Segments with low CV_w (typically $CV_w \leq 0.12$) exhibit very regular width profiles with a small amplitude of local widening and narrowing, and the melt-pool edges appear stable. These segments generally also pass the gap test and therefore correspond to optimum conditions in terms of surface tension, wetting behaviour and melt-pool feeding. Intermediate values ($0.12 < CV_w \leq 0.25$) indicate that width oscillation within the segment has started to increase; in these regions the change in layer thickness along the inclined bed has a pronounced influence on melt-pool behaviour. When CV_w exceeds approximately 0.25, strong irregularities, abrupt necking, local breaks and multiple micro-gaps are observed. In such segments the melt pool enters a highly unstable regime, and both continuity and geometric homogeneity are degraded.

3.1.3. Combined Classification

The combined (Gap + Uniformity) classification separates the segments into three main quality classes: stable, unstable and critical. Stable segments are characterised by both low gap length and low CV_w and represent the most favourable process conditions. These segments are most frequent for the (30 A, 2 mm/s) and (30 A, 3 mm/s) parameter combinations. Unstable segments are associated with intermediate L_{gap} and CV_w values; the track is still globally continuous, but stability is weakened. This behaviour is typically seen for (20 A, 2 mm/s) and (25 A, 3 mm/s). Critical segments exhibit simultaneously large gap lengths and high CV_w ; in these regions the melt pool repeatedly breaks up, the track is split into short pieces and the process becomes highly unstable. Critical behaviour clusters at high-speed, low-energy combinations such as (20 A, 4 mm/s), (25 A, 4 mm/s) and (30 A, 5 mm/s).

Overall, the segment-based analysis demonstrates that single-track PBAAM behaviour on the inclined bed cannot be represented by a single scalar quantity. A given track can contain stable, unstable and critical segments at different positions. This heterogeneity is particularly pronounced on the inclined bed, where the increasing layer thickness from left to right significantly alters melt-pool geometry, wetting performance and heat flow. Consequently, a physically meaningful evaluation of track quality requires a segment-wise approach. The proposed method not only provides a more accurate representation of continuity but also supports the construction of process windows, the identification of optimum parameter sets and the development of layer-dependent quality prediction models for PBAAM.

Based on the continuity and uniformity metrics summarized in Table 1, a thickness-resolved process window diagram was constructed (Figure 5), integrating current, travel speed, and the spatially varying local powder thickness along the inclined bed. Stability was defined using the CV derived from the continuity analysis, where low CV values represent stable formation and high CV values correspond to unstable regimes. This representation

provides a direct, engineering-oriented visualization of the parameter–thickness ranges that enable continuous and stable track formation.

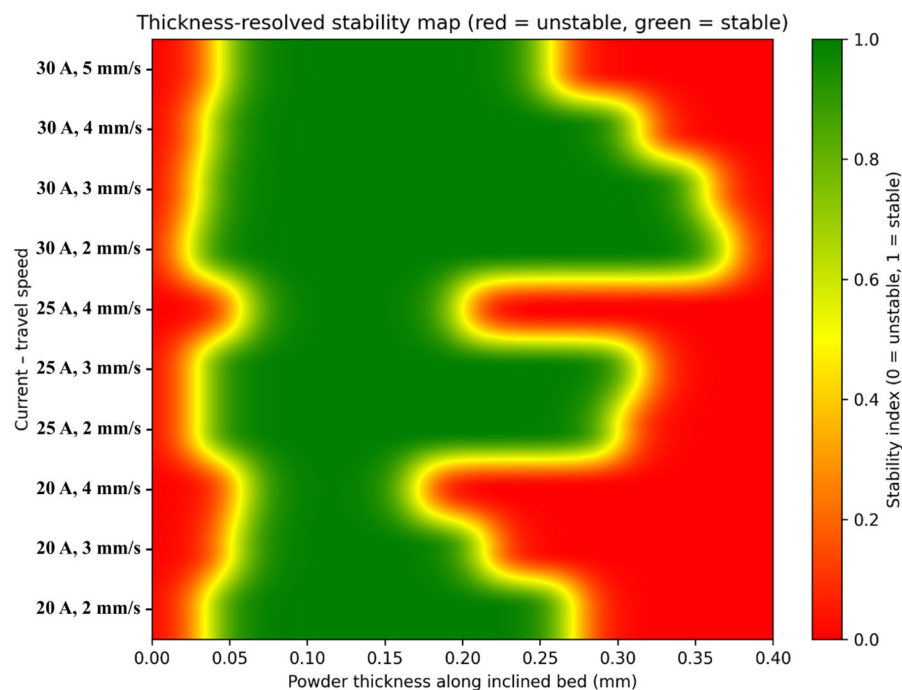


Figure 5. Thickness-resolved process window diagram based on CV-defined stability criteria.

3.2. Interpretation of Width-Profile Statistics on the Inclined Bed

The thickness (width) profiles confirm that melt-pool stability in single-track PBAAM is strongly governed not only by the process parameters but also by the layer-thickness variation along the inclined bed. An increase in current (20 → 25 → 30 A) leads to a clear increase in track width and a reduction in the coefficient of variation CV_w ; especially at 30 A, the higher heat input produces a wider and more stable melt pool. In contrast, increasing the travel speed (2 → 3 → 4 → 5 mm/s) weakens capillary feeding of the pool, narrows the profile, introduces multiple local minima and raises the CV_w values, which indicate a rapid deterioration in track stability.

The spatial change in layer thickness from left to right along the inclined bed has a pronounced effect on the width profiles for all parameter sets. In the thin-layer region near the track entrance, the width is more stable and typically forms a single, long segment. As the powder layer becomes thicker towards the right-hand side, melt-pool penetration decreases, the track narrows and the profile become more fragmented. Considering current, speed and layer thickness together, the most stable width profiles are obtained at high current (30 A) and low-to-moderate travel speeds (2–3 mm/s). In contrast, low-current/high-speed combinations combined with increasing layer thickness led to highly segmented, necking-prone profiles with large variance. These observations confirm that single-track PBAAM behaviour cannot be assessed solely from process parameters; the spatial evolution of layer thickness must be considered, and optimum stability is found where high current, low speed and the thin-layer region coincide.

3.3. Width Evolution Along the Inclined Track

Single-track PBAAM experiments performed at 20–30 A and 2–4 mm/s on the inclined powder bed (0–0.4 mm) further reveal how melt-pool width evolves along the scan direction (Figure 6). Overall, the melt-pool width increases from the entrance towards the middle of the track and, for some parameter sets, decreases again near the end. This behaviour is

attributed to changing heat-conduction conditions within the substrate–powder system as the powder layer becomes thicker, and to the resulting current–speed–thickness dependence of melt-pool stability.

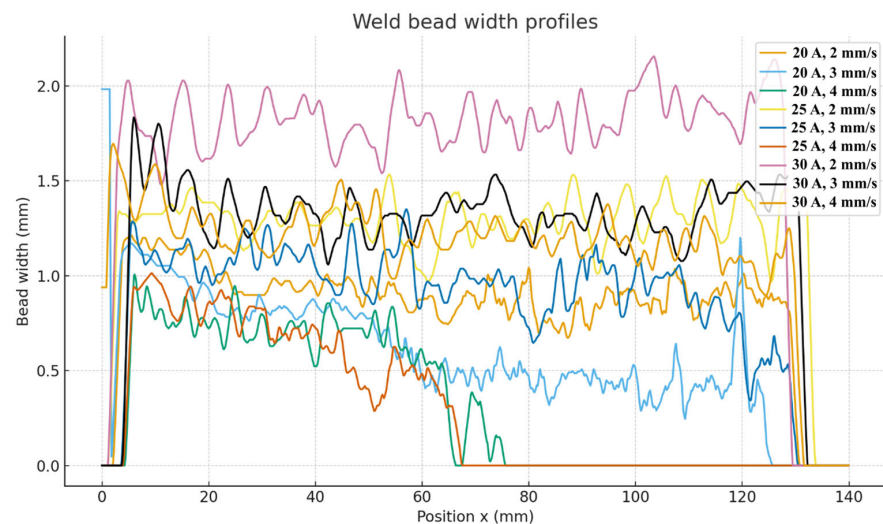


Figure 6. Evolution of melt-pool width along the inclined powder bed for different combinations of current (20–30 A) and travel speed (2–4 mm/s).

The melt pool width varies along the start, middle, and end positions of the inclined powder bed depending on the applied current and travel speed (Figure 7). At low current (20 A), initial widths are in the range of 2.10–2.46 mm; a moderate widening is observed towards the middle section, while the trailing region exhibits only limited narrowing in most cases. At medium current (25 A), starting widths of 2.20–3.65 mm are measured, with a more pronounced tendency to widen along the track and final widths reaching approximately 3.67–3.74 mm. At high current (30 A), melt-pool widths increase significantly: initial values of 3.66–4.83 mm are obtained, and maximum widths of about 4.20–4.75 mm occur in the mid-region. However, for the high-current/high-speed combination (30 A 4 mm/s), the width near the end of the track drops to around 3.26 mm, indicating that at high speed, the melt pool becomes less stable when combined with increased powder thickness.

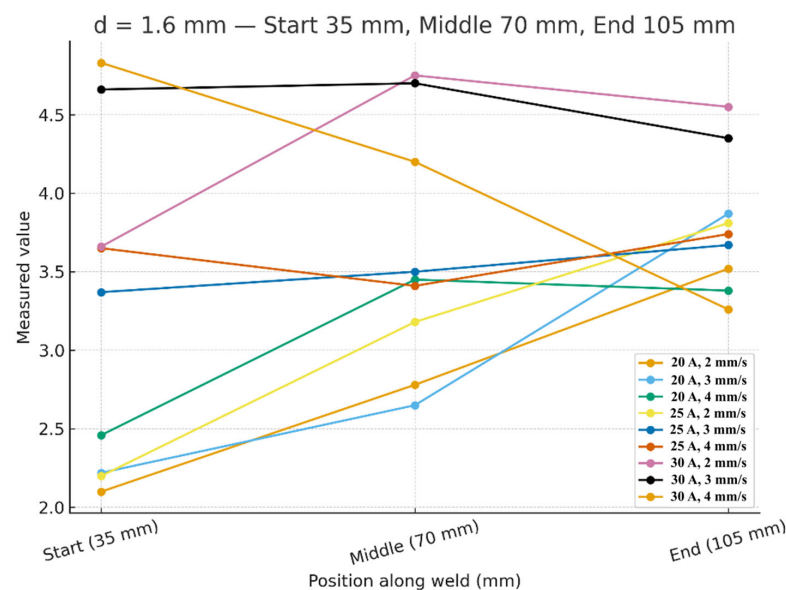


Figure 7. Evolution of melt-pool width along the inclined powder bed (start, middle and end positions) for different combinations of current and travel speed.

In summary, increasing current consistently enlarges the melt-pool width in all regions, and low travel speeds help maintain this width more uniformly. The maximum width in the middle of the inclined bed suggests that the increased powder height in this region restricts heat flow into the substrate and thereby sustains a broader melt pool. The narrowing observed near the end of the track, especially at high speeds, is attributed to insufficient heat accumulation and shorter arc exposure time over the thickest powder. Taken together, these results indicate that the (30 A, 2 mm/s) condition yields the widest and most stable melt pool on the inclined bed, whereas high-speed/high-current combinations reduce melt-pool stability. This confirms that, in inclined-bed PBAAM, both heat input and layer-thickness evolution directly control the melt-pool geometry along the track.

3.4. Intra-Layer Melting Experiments at Different Hatch Distances

Based on the inclined-bed experiments, it was concluded that the solidification geometries obtained with the 1.6 mm diameter tungsten electrode were the most stable and regular. Therefore, this electrode diameter was selected for the intra-layer hatch-spacing study, and process parameters were chosen by combining different travel speeds and layer thicknesses.

From the inclined-bed tracks produced with the 1.6 mm electrode, three layer-thickness regions were selected for detailed melt-pool characterization: 0.15 mm, 0.20 mm and 0.25 mm. For each layer thickness, a parameter set that exhibited stable solidification on the inclined bed was identified. Cross-sections were then taken from these regions to image the melt pool and quantify the penetration into the substrate. As shown in Figure 8, the positions used for sectioning correspond to (30 A 5 mm/s) for the 0.15 mm layer, (25 A 3 mm/s) for the 0.20 mm layer and (30 A 4 mm/s) for the 0.25 mm layer. Examination of the marked locations confirms that stable solidification zones are formed for these parameter sets.

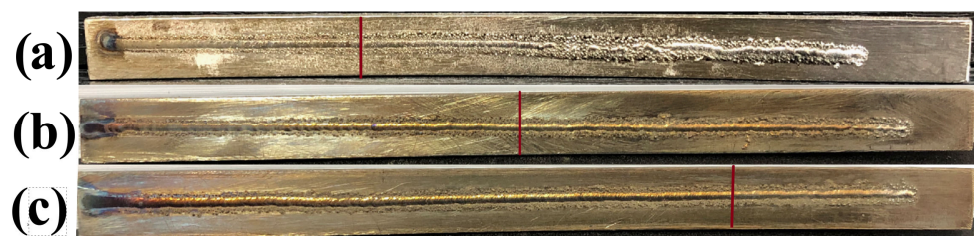


Figure 8. Schematic illustration of the selected cross-section locations on the inclined powder bed for layer thicknesses of (a) 0.15 mm, 30 A, 5 mm/s, (b) 0.20 mm, 25 A, 3 mm/s, and (c) 0.25 mm, 30 A, 4 mm/s. (Sections taken from regions marked by red lines).

The cross-sectional melt-pool images obtained from these regions are presented in Figure 9. The measured penetration widths at the substrate interface were 0.88 mm for 0.15 mm layer thickness at (30 A 5 mm/s), 1.25 mm for 0.20 mm layer thickness at (25 A 3 mm/s), and 1.07 mm for 0.25 mm layer thickness at (30 A 4 mm/s). Using these measured widths as reference, hatch distances were defined as 25%, 50% and 75% of the melt-pool width, in line with typical overlap ratios reported in the LPBF literature. For each layer thickness, the corresponding hatch distances were obtained by multiplying the measured penetration width by the selected percentage and used as the centre-to-centre spacing between two adjacent melt tracks.

The solidified top views obtained after applying different hatch distances are shown in Figure 10. As the hatch distance increases, the overall width of the linear solidification region increases as well. For hatch ratios of 25% and 50%, the top-view geometries appear as nearly single-line tracks with strong overlap between neighbouring passes. At a hatch

ratio of 75%, however, the geometry follows the arc trajectory in a clearly double-lined manner, and irregularities are observed in the arc start and end regions. For all hatch ratios, the solidification remains essentially linear, and cross-sections were taken from the mid-length of the melted regions to assess melt-pool penetration and inter-track fusion.

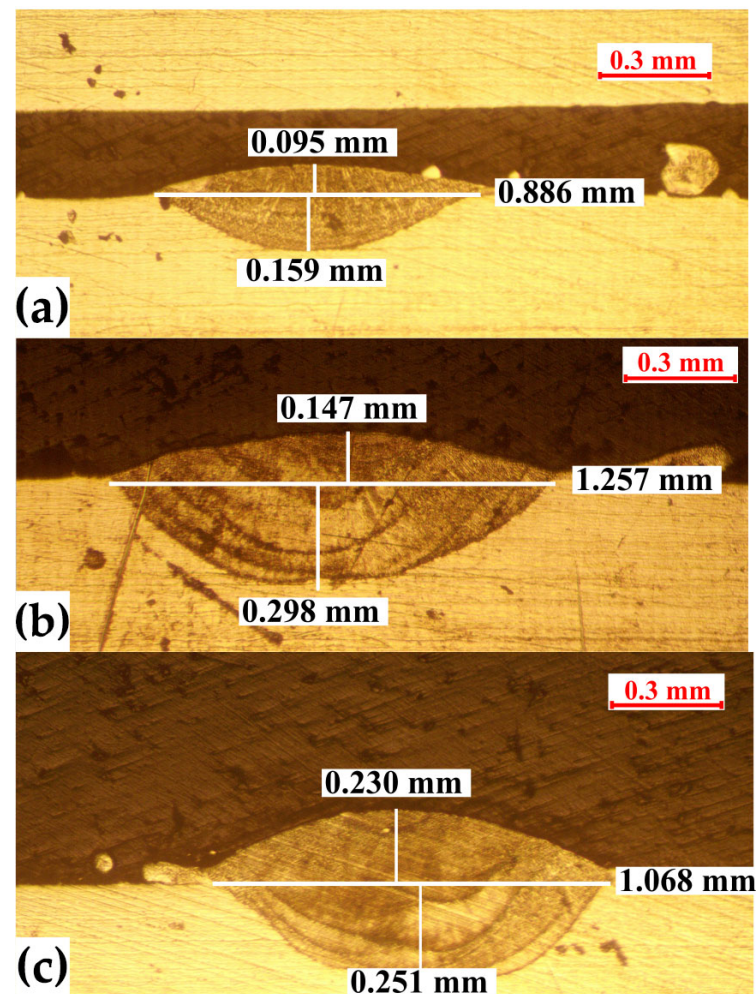


Figure 9. Cross-sectional melt-pool geometries at the selected inclined-bed locations for layer thicknesses of (a) 0.15 mm (30 A, 5 mm/s), (b) 0.20 mm (25 A, 3 mm/s), and (c) 0.25 mm (30 A, 4 mm/s).

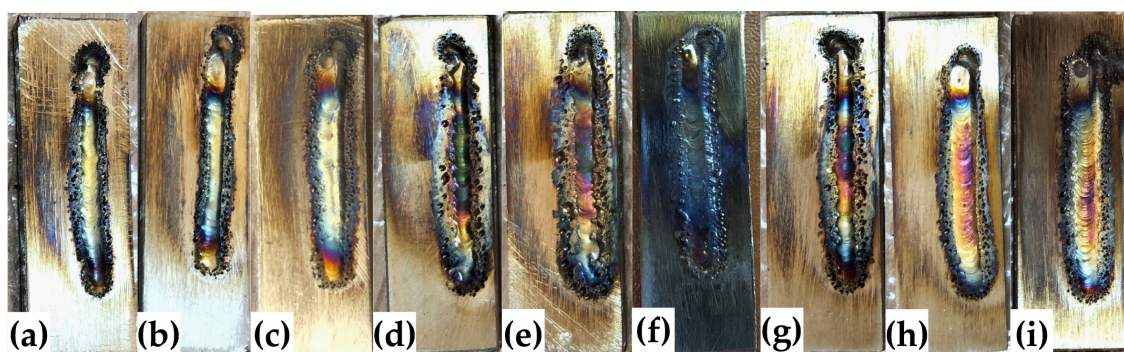


Figure 10. (a–c) 0.15 mm layer thickness at 30 A 5 mm/s with hatch distances of 25%, 50%, and 75% of the melt-pool width, respectively. (d–f) 0.20 mm layer thickness at 25 A 3 mm/s with 25%, 50%, and 75% hatch spacing. (g–i) 0.25 mm layer thickness at 30 A 4 mm/s with 25%, 50%, and 75% hatch spacing.

Intra-layer hatch-spacing conditions were then evaluated in terms of thickness (melt-pool) profile, coefficient of variation CV_w , connected-component analysis, skeleton-based continuity index CL , and gap-based fusion criteria (Table 2). The results show that at high current levels (the 30 A parameter group), the tracks form a single connected component for all hatch ratios, the skeleton-based continuity index reaches $CL = 1.00$, and the minimum inter-track gap drops to zero. This indicates that the high energy input produces robust melt-pool geometries and guarantees track continuity.

Table 2. Summary of hatch-spacing experiments: average melt-pool thickness, width stability and continuity indices.

Parameter Set	Hatch (%)	Mean Thickness (mm)	CV_w	Continuity	CL	Assessment
30 A, 5 mm/s	25	0.612	0.455	Yes	1.00	Wide track good fusion
30 A, 5 mm/s	50	0.731	0.487	Yes	1.00	Deep and wide melt pool
30 A, 5 mm/s	75	0.534	0.498	Yes	1.00	Narrowing pool continuity kept
25 A, 3 mm/s	25	0.485	0.412	Yes	1.00	Stable narrow track
25 A, 3 mm/s	50	0.503	0.365	Yes	1.00	Most stable profile
25 A, 3 mm/s	75	0.391	0.512	No	0.43	Discrete track insufficient energy
30 A, 4 mm/s	25	0.558	0.441	Yes	1.00	Wide and stable track
30 A, 4 mm/s	50	0.614	0.473	Yes	1.00	Strong fusion
30 A, 4 mm/s	75	0.462	0.495	Yes	1.00	Narrow track not recommended for multi-layer builds

Under low-current/high-hatch conditions, especially for the (25 A, 3 mm/s, 75%) hatch parameter set, the melt track splits into two components, the gap value remains above the critical threshold (defined as $0.5 w_s$), and the skeleton-based CL decreases to 0.43. According to the continuity criteria, this condition is therefore classified as a *discrete track* with insufficient fusion between neighbouring passes.

Another important finding is obtained for the (30 A, 4 mm/s, 75% hatch) condition: although the continuity is preserved ($CL = 1$), the track width becomes significantly narrower. This “thin-track” morphology is associated with increased lateral energy spread, and from a multi-layer PBAAM/LPBF perspective it is not considered suitable, because it cannot provide a sufficiently large fusion surface for bonding to subsequent layers. Previous studies have reported that excessively narrow tracks lead to poor interlayer bonding, increased interfacial stresses and a higher risk of delamination in multi-layer builds [22].

Thickness-profile analysis shows that the highest average melt-pool thickness (or effective cross-sectional width) of 0.731 mm is obtained for the (30 A, 5 mm/s, 50% hatch) condition. According to the CV_w values, the most stable track geometry is achieved for the (25 A, 3 mm/s, 50% hatch) parameter set. Overall, the results demonstrate that changes in energy density and hatch spacing significantly affect single-track morphology and continuity behaviour.

Representative melt-pool regions for different hatch ratios are presented in Figure 11, highlighting the differences in penetration depth, fusion between neighbouring tracks and solidification structure.

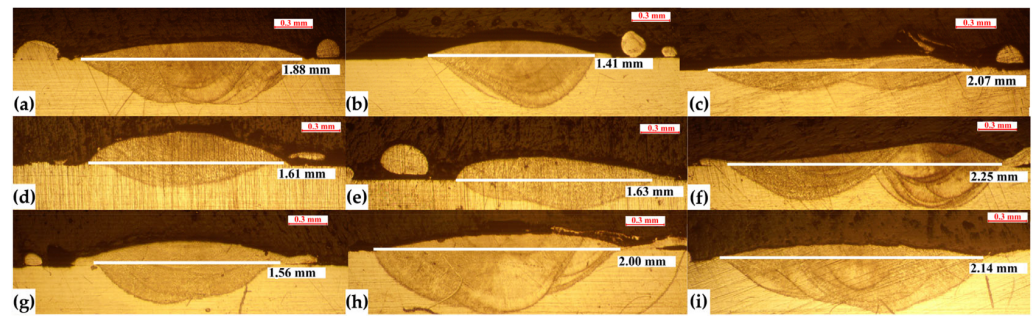


Figure 11. Cross-sectional morphology of adjacent melt tracks produced under different layer thickness and hatch distance conditions. (a–c) 0.15 mm layer thickness at (30 A, 5 mm/s) with hatch distances corresponding to 25%, 50%, and 75% of the measured melt-pool width, respectively. (d–f) 0.20 mm layer thickness at (25 A, 3 mm/s) with 25%, 50%, and 75% hatch spacing. (g–i) 0.25 mm layer thickness at (30 A, 4 mm/s) with 25%, 50%, and 75% hatch spacing.

3.5. Multi-Layer Melting Experiments

The intra-layer hatch-spacing and inclined-bed results showed that a hatch ratio of 75% is insufficient to ensure linear penetration and bonding between neighbouring tracks. At 75% hatch, the lack of overlap leads to unmelted or only partially sintered regions on top of the solidified layer, preventing the formation of a continuous molten domain. For this reason, multi-layer melting experiments were carried out only with 25% and 50% hatch ratios at the three layer-thickness parameter sets selected from the inclined-bed study. The corresponding solidified surface morphologies are presented in Figures 12–14. In all cases, the number of deposited layers was chosen such that the total built height was at least 1 mm of powder.

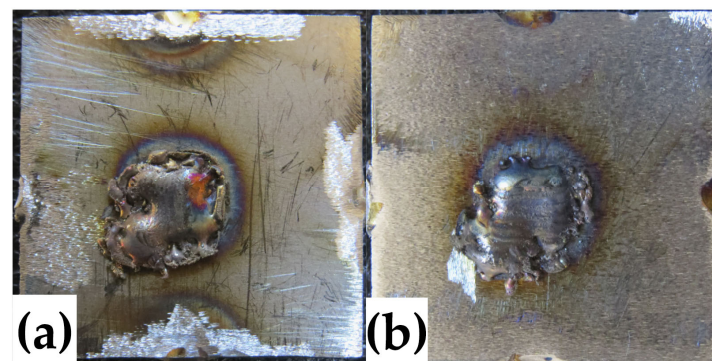


Figure 12. Multi-layer solidification morphologies for a layer thickness of 0.15 mm at 30 A and 5 mm/s with hatch ratios of (a) 25% (0.22 mm) and (b) 50% (0.44 mm)/0.66 mm.

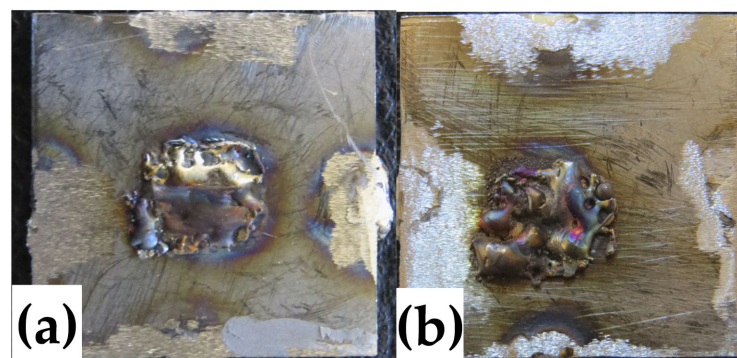


Figure 13. Multi-layer solidification morphologies for a layer thickness of 0.20 mm at 25 A and 3 mm/s with hatch ratios of (a) 25% (0.31 mm) and (b) 50% (0.63 mm).

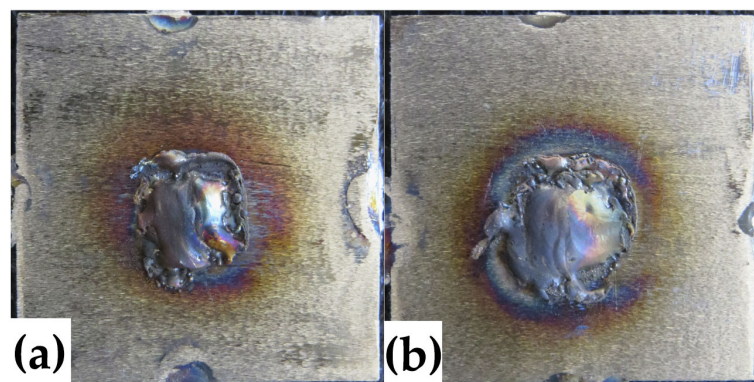


Figure 14. Multi-layer solidification morphologies for a layer thickness of 0.25 mm at 30 A and 4 mm/s with hatch ratios of (a) 25% (0.27 mm) and (b) 50% (0.54 mm).

For all multi-layer tracks, irregularities were observed near the arc starting locations, and these irregularities manifested as localized balling. In multi-layer arc-based additive manufacturing processes, progressive thermal accumulation significantly influences melt-pool geometry, dimensional accuracy and residual stress development. As successive layers are deposited, the temperature gradient between the newly added material and the underlying substrate decreases, leading to reduced heat dissipation efficiency and higher local heat retention. This may result in geometric distortion, melt-pool widening, and the development of residual stresses due to constrained thermal contraction [22].

During the present five-layer experiments, minor geometric deviations were observed under higher heat-input conditions, particularly in low-speed/high-current combinations. Localized lack-of-fusion features were also detected in limited regions. These observations are consistent with the progressive heat accumulation mechanisms reported in arc-based additive manufacturing systems [23]. However, despite these localized effects, stable stacking of five consecutive layers was achieved within the investigated parameter window. A comprehensive thermo-mechanical assessment, including quantitative residual stress measurement and porosity distribution analysis across extended builds, is beyond the scope of the present study. In Figure 12, which corresponds to a layer thickness of 0.15 mm (30 A current, 5 mm/s travel speed), the 25% hatch condition (0.22 mm spacing) exhibits pronounced balling and localised energy accumulation in the arc region. When the hatch ratio is increased to 50% (0.44 mm spacing), the solidified geometry becomes more regular, except at the track initiation point, and the melt tracks more closely match the desired linear geometry. In addition, secondary remelted zones and sintered powder regions are detected in the heat-affected areas outside the main melt track.

In Figure 13, corresponding to a layer thickness of 0.20 mm (25 A, 3 mm/s), a 25% hatch ratio (0.31 mm spacing) is sufficient to maintain melt continuity; however, increased balling and irregularities are observed at the track edges due to the thicker powder layer. When the hatch ratio is increased to 50% (0.63 mm spacing), the local energy density becomes inadequate, the molten region becomes highly irregular and the overall geometry degrades severely.

The solidified geometries for the 0.25 mm layer thickness (30 A, 4 mm/s) are shown in Figure 14. At 25% hatch (0.27 mm spacing), excessive energy input causes significant convexity of the surface (“dome” formation) and an increase in build height. For the 50% hatch condition (0.54 mm spacing), the dome height decreases but peripheral balling becomes more pronounced. No multi-layer experiments were performed at 75% hatch for any of the layer thicknesses, owing to the lack of sufficient overlap observed in the intra-layer tests.

To analyse the melting behaviour within the stacked layers, cross-sections were taken from the multi-layer builds. The cut samples were mounted in epoxy, ground and polished, and then etched to reveal the melt-pool cross-sections. The corresponding melt pool images for the different hatch ratios and layer thicknesses are shown in Figures 15–17.

Examination of the multi-layer cross-sections shows that in Figure 15—particularly at the 25% hatch ratio—the melt pools exhibit a pronounced keyhole geometry. In contrast, Figures 16 and 17 mostly show conduction-mode melting behaviour. Melt-pool dimensions measured from these cross-sections are summarised in Table 3. Using these experimental results, the influence of hatch ratio and layer thickness on melt-pool geometry is discussed in detail and compared with theoretical and experimental data reported in the literature, to elucidate process–microstructure relationships.

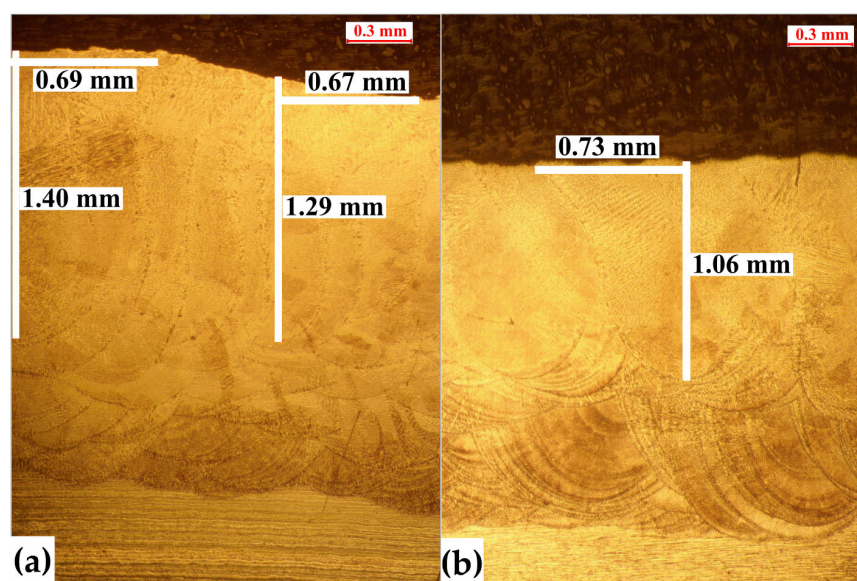


Figure 15. Cross-sectional microstructures for a layer thickness of 0.15 mm at 30 A and 5 mm/s with hatch ratios of (a) 25% (0.22 mm) and (b) 50% (0.44 mm).

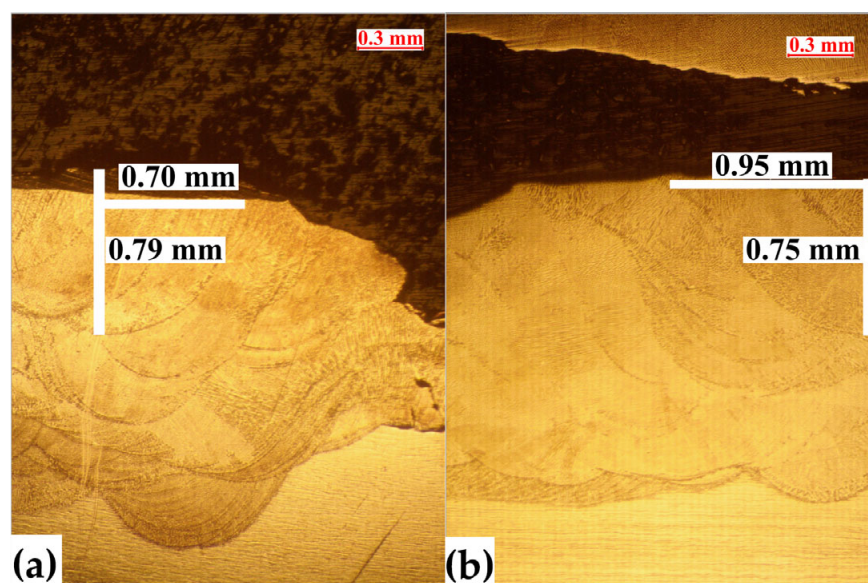


Figure 16. Cross-sectional microstructures for a layer thickness of 0.2 mm at 30 A and 5 mm/s with hatch ratios of (a) 25% (0.27 mm) and (b) 50% (0.54 mm).

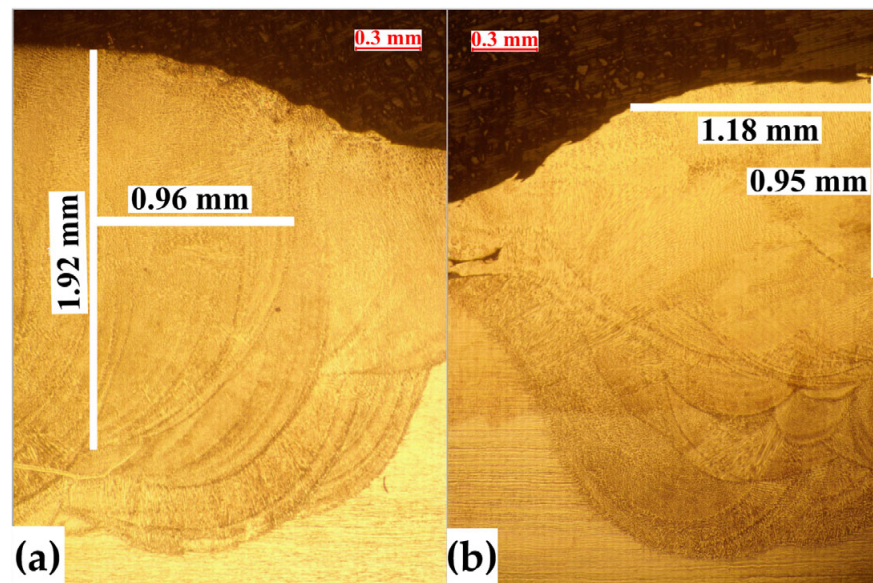


Figure 17. Cross-sectional microstructures for a layer thickness of 0.25 mm at 30 A and 4 mm/s with hatch ratios of (a) 25% (0.31 mm) and (b) 50% (0.63 mm).

Table 3. Melt-pool dimensions for multi-layer PBAAM experiments.

Layer Thickness (mm)	Current (A)	Travel Speed (mm/s)	Hatch Ratio	Hatch Distance (mm)	Melt-Pool Depth (mm)	Melt-Pool Width (mm)
0.15	30	5	25%	0.22	1.29	0.67
			50%	0.44	1.06	0.73
0.20	25	3	25%	0.27	0.79	0.70
			50%	0.54	0.75	0.95
0.25	30	4	25%	0.31	1.92	0.96
			50%	0.63	0.75	1.18

In all parameter combinations, increasing the hatch ratio from 25% to 50% leads to a decrease in melt-pool depth and a simultaneous increase in melt-pool width. This behaviour is attributed to the inverse relationship between energy density and hatch distance. A larger hatch spacing reduces the energy delivered per unit area, limiting the depth of melting, while the thermal field is distributed over a wider lateral region, producing a shallower but broader melt pool. Schwerz and Nyborg (2021) similarly reported that scan parameters strongly control melt-pool dimensions and that low overlap ratios (i.e., high hatch distances) promote lack-of-fusion defects [24]. This trend is further supported by the review of Wang et al. (2023), who showed that hatch spacing and travel speed directly affect melt-pool stability and shape; at large hatch distances, melt pools tend to become shallow and laterally extended [25].

With respect to melt-pool width, the present results reveal a clear increasing trend with increasing hatch ratio. The maximum width of 1.18 mm is obtained at a layer thickness of 0.25 mm and a 50% hatch ratio. This is consistent with the observations of Wang et al. (2023), who reported up to 100% increases in melt-pool width when the interlayer temperature was raised from 100 °C to 300 °C, due to thermal accumulation effects [25]. Such heat accumulation favours the formation of wide but shallow melt pools, which in turn influences remelting behaviour and microstructural homogeneity.

Overall, the data in Table 3 demonstrate that both hatch ratio and layer thickness govern the balance between melt-pool depth and width in multi-layer PBAAM. Low hatch spacing (25%) promotes deeper penetration and, in some cases, keyhole-mode melting, whereas higher hatch spacing (50%) favours wider, shallower pools and increases the risk of insufficient interlayer fusion if the energy input is not properly adjusted.

4. Discussion

4.1. Evaluation of Thickness Profiles on the Inclined Bed

The results of this study show that the melt-pool width is highly sensitive not only to the combination of current and travel speed, but also to the variation in powder layer thickness along the inclined bed. The marked increase in melt-pool width at all measurement points when the current is raised from 20 A to 30 A confirms the well-known positive correlation between heat input and melt-pool size reported in additive manufacturing studies [22]. The fact that the initial widths exceed 4 mm at 30 A indicates that the same mechanism is valid for TIG-based powder-bed arc additive manufacturing (TIG-PBAAM).

However, the melt pool is not governed by current alone; it is also shaped by the change in thermal resistance associated with increasing powder layer thickness. Along the inclined bed, the layer thickness increases from 0 to 0.4 mm, which restricts heat conduction and promotes a shallower yet wider melt pool in the mid-region. This behaviour is consistent with experimental and computational studies showing that powder layers alter energy absorption and heat distribution compared with bare substrates [26,27].

Near the end of the track, especially at high current–high speed combinations, the melt-pool width decreases again. This can be attributed to the reduced interaction time of the arc with the substrate at high travel speeds. King et al. (2014) reported that high scan speeds force the melt pool into a longer but narrower morphology and that feed–heat imbalance can cause edge constriction [28]. In the present study, the width at 30 A–4 mm/s decreases from 4.20 mm in the mid-region to 3.26 mm in the trailing region, which is consistent with this description.

The observed widening–narrowing cycle along the inclined bed is also linked to heat accumulation during continuous scanning. Rometsch et al. (2022) showed that the melt pool often reaches its maximum width in the mid-section of a track and then narrows as the heat-flow paths evolve [29,30]. In our results, the maximum width occurs in the middle and a partial narrowing appears towards the end, particularly at high speeds, supporting this picture. By contrast, high-heat-input/low-speed conditions such as 30 A–2 mm/s generate a wide and stable melt pool along the entire track, in line with the “wide pool = improved wetting and fusion quality” relation reported by Zhao et al. (2017) [31].

In summary, the TIG-PBAAM experiments conducted with a 1.6 mm electrode on the inclined powder bed demonstrate that melt-pool geometry is controlled not only by process parameters, but also by bed geometry and layer thickness. While increasing current enlarges the pool in all regions, the growing powder thickness in the mid-region promotes a wide surface-confined pool, and high-speed leads to narrowing in the trailing region. For inclined beds, optimum fusion therefore requires combining high current with low travel speeds, particularly where the layer thickness—and thus thermal resistance—is elevated.

4.2. Continuity Behaviour of Single Tracks on the Inclined Bed

The updated continuity analysis shows that single-track PBAAM rarely forms a single perfectly continuous bead; instead, the track is usually composed of multiple continuity segments. For medium-to-low heat input, repeated local necking and breakup events occur along the track, dividing it into several continuous intervals. Similar segmental behaviour

has been reported in LPBF and arc-based AM, where “stability bands” form because of local thermal gradients, powder-bed height variations and fluctuations in wetting angle [29,30].

The gap data highlight the key role of the current–speed interaction in controlling track integrity. For 25 A 4 mm/s and 20 A 4 mm/s, the total gap length L_{gap} rises to 77–81 mm, indicating that at high scan speeds the melt pool cannot be sufficiently fed along the track and a breakup-dominated regime is established. This agrees with reports that high scan speeds in arc-based additive manufacturing cause intermittent melt-pool formation and long discontinuity segments that degrade wetting quality [30]. In contrast, when the current is 30 A and the speed is kept in the 2–3 mm/s range, L_{gap} is only about 13 mm and the gaps are mainly confined to the start–end regions. These findings are consistent with directed energy deposition (DED) studies that link increased energy input to improved melt-pool stability and reduced porosity [32].

The uniformity metric CV_w exhibits a strong parallel with continuity. Tracks with low CV (approximately $CV_w \leq 0.12$) show limited width fluctuation and long, single-piece continuity segments. For example, for 30 A 2 mm/s ($CV = 0.085$) and 30 A 3 mm/s ($CV = 0.114$), the width variation along the track is modest and the bead behaves largely as one stable segment. In contrast, highly variable tracks such as 20 A 3 mm/s ($CV = 0.612$) contain many local necking regions and are split into multiple segments. These findings are consistent with reports where width-variation or geometry-variability metrics are used as predictors of melt pool stability and lack-of-fusion behaviour [20,33].

Threshold selection and validation. In addition to reporting the cumulative gap length along the track, the continuity decision was based on a local gap-length criterion, L_{gap} , to reflect the geometric scale of arc-based tracks. The adopted threshold $L_{crit} = 0.5$ mm corresponds to ~15–25% of the typical PBAAM track width (~2–4 mm), following the same ratio-based reasoning used in single-track continuity assessments, while being proportionally larger than thresholds used in finer-scale laser processes [34]. For the uniformity criterion, prior studies commonly associate stable single-track behaviour with CV_w values in the ~0.10–0.15 band [35]; therefore, a sensitivity scan was performed within this range, and the highest agreement with expert visual top-view labels was obtained at $CV_{crit} = 0.12$. The selected pair ($L_{crit} = 0.5$ mm, $CV_{crit} = 0.12$) was validated against repeated measurements and visual labeling across the full parameter space, yielding ~95–97% agreement, supporting its suitability for the present PBAAM setup. A conservative combined rule was used: a track is classified as discontinuous if ($L_{gap} > 0.5$ mm) or ($CV_{crit} \geq 0.12$), and as continuous only if both criteria are satisfied. Literature from LPBF is cited only as a ratio-level sanity check (e.g., CV ranges reported in [36]) and as context for single-track criteria discussions, rather than as a direct source of the thresholds.

When the strict global criterion is applied over the entire geometric track length, no parameter combination satisfies both gap and uniformity thresholds simultaneously. However, when the evaluation is restricted to the active continuity interval, high-current/low-speed combinations (notably 30 A, 2 mm/s and 30, 3 mm/s) satisfy both criteria within extended spatial regions. These gaps correspond to the well-known “lead-in/lead-out thermal transients” in AM tracks, which arise from initial heating and final cooling effects [37]. In the revised framework, these transient regions are formally excluded from the quantitative assessment. Instead, evaluation should focus on the active continuity interval and segment stability. On the inclined bed, the layer thickness increases from left to right, so the bead geometry naturally evolves along the track. For this reason, it is more meaningful to assess continuity and uniformity using segment-based physical interpretation rather than a single absolute threshold [20].

4.3. Effect of Different Hatch-Distance Values

Single-track experiments with different hatch ratios clearly demonstrate the link between energy density and fusion behaviour. Under low energy input, particularly at (25 A, 3 mm/s, 75% hatch), a pronounced lack of fusion is observed, characterised by loss of continuity, increased gap length and a skeleton-based continuity index CL dropping to 0.43. This behaviour is consistent with studies showing that low energy density and excessive hatch spacing prevent adequate wetting and bonding between neighbouring tracks [21].

The beneficial effect of high current on continuity agrees with the mechanism described by DebRoy et al. (2018), who reported that increased energy input enhances melt-pool stability [22]. A notable finding of the present work, however, is that at (30 A, 4 mm/s, 75% hatch), the track remains continuous but becomes geometrically very narrow. In such narrow tracks, lateral energy spreading increases, pool width decreases and the contact area available to interact with the overlying layer is reduced. As a result, narrow beads are prone to cold bonding, insufficient fusion, interfacial delamination and accumulation of thermal stresses in multi-layer builds [22,37]. This parameter set should therefore be avoided in practical multi-layer PBAAM.

Overall, the combined use of the skeleton-based continuity index CL gap analysis and connected-component testing enables an objective and repeatable classification of single-track fusion quality. This approach is in line with recent work proposing geometric and statistical track-assessment methods [32,38] and provides a robust decision framework for hatch optimisation and the design of multi-layer strategies.

4.4. Melt Mode and Pool Dimensions in Multi-Layer Melting

The melt-pool shapes observed in the multi-layer experiments exhibit the typical characteristics of conduction and keyhole regimes. In conduction mode, the energy is primarily absorbed at the surface and transported to the underlying material by conduction, producing shallow, semi-circular melt pools. As the energy density increases, a vapour channel forms within the liquid metal, giving rise to deeper, narrower keyhole pools and an elevated risk of porosity [38]

In the present work, cross-sections at 0.15 mm layer thickness with 25% hatch show pronounced keyhole features, whereas the other parameter combinations exhibit mostly conduction-mode melting. The data demonstrates that increasing the hatch ratio from 25% to 50% reduces melt-pool depth while increasing its width for all conditions. This trend reflects the inverse relationship between energy density and hatch spacing and the direct relationship between lateral heat-affected area and pool width [38]

Layer thickness and travel speed also play a decisive role in melt-pool depth. At 0.25 mm layer thickness, the combination of 30 A 4 mm/s–25% hatch produces a maximum depth of 1.92 mm, indicating that thermal accumulation at low speed becomes particularly pronounced in thick layers. This observation is consistent with multiphysics simulations and DED experiments showing deeper pools at low scan speeds due to increased energy input per unit length [32]. In contrast, at 0.20 mm layer thickness, the lower current of 25 A results in shallow conduction-mode pools with depths of 0.75–0.79 mm, in line with the formation of shallow, unstable pools under low energy density [38].

These results suggest that in multi-layer PBAAM, the melt mode and pool dimensions must be adjusted deliberately with respect to layer thickness and hatch spacing. Otherwise, excessive energy density may trigger keyhole-related porosity, while insufficient energy may lead to lack-of-fusion defects.

4.5. Overall Assessment of the Powder-Bed Arc Melting Process

This study has systematically investigated the influence of process parameters (current, travel/scan speed, layer thickness and hatch distance) on melt-pool geometry, track continuity and multi-layer stability in a powder-bed arc additive manufacturing (PBAAM) system. Experiments with 316L stainless steel powder demonstrate that, within suitable parameter windows, this process can produce continuous, metallurgically bonded structures.

The literature indicates that arc-based systems provide wider melt pools and higher deposition rates than LPBF, but at the cost of reduced geometric accuracy [2,30]. The present work shows that single-track pool widths are in the millimetre range and that the optimum layer thickness is approximately 0.3–0.5 mm. This places PBAAM between LPBF (30–60 μm) and WAAM (1.5–2 mm) in terms of resolution and layer size [29,39].

As current increases, both melt-pool width and penetration depth increase, whereas increasing travel speed reduces the energy per unit length and produces narrower, shallower pools. This trend is consistent with power–speed–pool-size relationships reported for LPBF and laser metal deposition (LMD) [40]. In single-layer trials, when the hatch spacing is smaller than the melt-pool width, neighbouring tracks overlap sufficiently, and continuous fusion is achieved. When the hatch spacing approaches or exceeds the width, unmelted regions and gap-based discontinuities appear between tracks. At very small hatch values (high overlap), excessive superposition leads to surface undulations and ridges, like the “swell-undercut” defects associated with excessive energy density.

In multi-layer builds, interlayer heat accumulation is particularly significant at high-current/low-speed combinations, where the track width increases with the number of layers and a conical expansion of the upper part of the build is observed. This resembles the distortion and loss of dimensional accuracy reported in WAAM [39]; however, because PBAAM uses smaller melt pools and thinner layers, the magnitude of these deviations is smaller. Offset hatch strategies help reduce peak–valley patterns by enabling upper layers to partially fill the valleys of the previous layer, like common scanning strategies in LPBF.

Beyond geometric widening, progressive thermal accumulation in multi-layer arc-based additive manufacturing significantly influences dimensional accuracy and residual stress evolution. As deposition height increases, reduced heat dissipation into the substrate modifies thermal gradients and cooling rates, which directly affect distortion and stress build-up due to constrained thermal contraction [41]. Thermo-mechanical modelling studies have shown that insufficient interlayer cooling leads to measurable geometric deviation and residual stress accumulation as layer number increases [42]. Experimental investigations on arc-based systems further confirm that heat build-up governs distortion behaviour and structural stability in multi-layer builds [43]. Although residual stresses were not quantitatively measured in the present study, the observed conical expansion and minor geometric deviations under high heat-input conditions are consistent with the thermo-mechanical behaviour reported in the literature.

From a metallurgical standpoint, appropriate parameter sets yield melt-pool depths exceeding the powder layer thickness, ensuring some degree of remelting beneath each layer and promoting strong interlayer bonding. In the inclined-bed experiments, when the effective powder thickness exceeds the pool depth, track continuity is lost and lack-of-fusion defects form [40].

Main Strengths of PBAAM:

- High deposition rate and efficiency, owing to relatively thick layers and larger melt pools compared with LPBF.
- Lower equipment cost and simpler infrastructure than laser or electron-beam systems, making it attractive for SMEs and emerging economies.

- Material flexibility, since the powder-bed configuration allows for different metal powders and metal–ceramic mixtures, enabling composite and functionally graded structures.
- Potential for strong metallurgical bonding and high density, provided that the process parameters are adequately optimised.

Key Limitations:

- Moderate geometric accuracy and surface roughness, with dimensional tolerances in the millimetre range.
- Risk of heat accumulation and distortion in multi-layer builds if cooling times are insufficient.
- A relatively narrow process window, where lack-of-fusion defects occur at low energy input and spatter or surface degradation appear at high energy input;

Limited microstructural and mechanical data for this process, which constrains the assessment of its industrial applicability [16,44].

- More detailed characterisation of microstructure and mechanical performance is therefore essential in future work.

5. Conclusions

This study investigated the influence of process parameters on melt-pool geometry, track continuity and multi-layer stability in TIG-based powder-bed arc additive manufacturing (PBAAM) of 316L stainless steel. The main findings can be summarised as follows:

- Inclined-bed experiments revealed that melt-pool width is strongly controlled by the interaction of current, travel speed and layer-thickness variation. Increasing current enlarges the pool in all regions, increasing layer thickness promotes a wide, surface-confined pool in the mid-region, while high speed causes narrowing near the track end.
- Single-track continuity analysis showed that tracks generally consist of multiple segments rather than one perfectly continuous bead. When evaluated within the active melting interval, high-energy parameter sets exhibit extended stable regions exceeding 100 mm along the inclined bed, corresponding to a broad powder thickness range. Width-based gap and uniformity metrics (L_{gap} and CV_w) reliably captured melt-pool stability: high-energy (30 A, 2–3 mm/s) conditions produced low gap and low CV values with stable segments, whereas low-energy /high-speed combinations led to discontinuous, high-variance tracks.
- Hatch-distance and intra-layer experiments demonstrated that hatch spacing plays a critical role in fusion behaviour. Hatch ratios of 25–50% provided sufficient overlap between neighboring tracks, while 75% hatch produced pronounced lack-of-fusion and narrow-track issues in both single-layer and multi-layer builds. The skeleton-based continuity index CL, combined with gap analysis, enabled objective classification of fusion quality.
- Multi-layer melting and cross-sectional analysis showed that 25% hatch conditions can produce keyhole-like deep pools for some parameter sets, whereas 50% hatch favours shallower but wider pools. The ratio between melt-pool depth and layer thickness was found to be crucial for full interlayer fusion; for the parameter sets considered here, the optimum layer-thickness range was approximately 0.3–0.5 mm.
- From a process-positioning perspective, PBAAM occupies an intermediate space between LPBF and WAAM in terms of melt-pool size and layer thickness, offering millimetre-scale resolution and high deposition rates suitable for medium-scale components. With appropriate parameters, crack-free and macroscopically pore-free structures with metallurgical bonding were obtained.

- By combining the inclined-bed approach, segment-based continuity analysis (CV_w , CL) and hatch–layer-thickness interaction, this work provides an experimental process map for PBAAM and a foundation for future optimisation and modelling efforts

Future work should extend the process window via design-of-experiments studies, integrate in situ monitoring and closed-loop control, systematically link microstructure to mechanical properties, explore multi-material and functionally graded structures, and develop hybrid process and numerical modelling approaches. These efforts will be essential for establishing PBAAM as a robust alternative metal additive manufacturing technology in both academic and industrial contexts.

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Abbreviations

The following abbreviations are used in this manuscript:

AM	Additive manufacturing
PBF	Powder bed fusion
LPBF	Laser powder bed fusion
SLM	Selective laser melting
EBM	Electron beam melting
WAAM	Wire arc additive manufacturing
PBAAM	Powder-bed arc additive manufacturing
TIG	Tungsten inert gas
AC	Alternating current
DC	Direct current
HD	Hatch distance
CV	Coefficient of variation
CL	Connected-length ratio
ND	Neutral density
RGB	Red-green-blue

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